

Computer Science & Information Technology

Computer Organization & Architecture

DPP 01

IO Organization

- Q1** 8-bit characters are transmitted using a synchronous mode of transmission with 1 start bit, 8 data bits and 1 stop bit. The efficiency of the transmission line is ____?
- Q2** 8-bit characters are transmitted using a parity synchronous mode of transmission with 1 start bit, 8 data bits, 2 stop bits and 1 parity bit. If the transfer rate of the line is 3000 bits per second, then effective transfer rate is _____ bytes per second?
- Q3** Consider a CPU which takes 0.05 microseconds as interrupt overhead time when a device generates interrupt for CPU, and CPU accepts it. After that CPU takes 5 cycles to service the interrupt. If CPU runs on 10MHz clock rate then total time CPU spends for interrupt service is _____ microseconds (rounded upto 2 decimal places)?
- Q4** Which of the following is connected to CPU directly?
 (A) Keyboard (B) Hard-disk
 (C) RAM (D) Camera
- Q5** Which of the following is/are true?
 1. Data format used in IO devices may differ from CPUs format
 2. IO devices are slower than CPU
 3. IO devices are slower than main memory
 (A) Only 1
 (B) Only 1 & 2
 (C) Only 1 & 3

(D) All 1, 2 & 3

Q6

Which of the following is true regarding IO mapped IO as compared to memory mapped IO?

1. ALU operation cannot be performed on IO data directly
 2. IO devices have their own address space
 3. Less number of Instructions to access IO
 4. Less number of IO devices connected
- (A) Only 2 & 3
 (B) Only 2 & 4
 (C) Only 2, 3 & 4
 (D) All 1, 2, 3 & 4

Q7

Which of the following is true regarding memory mapped IO as compared to IO mapped IO?

1. ALU operation cannot be performed on IO data directly
 2. IO devices do not have their own address space
 3. Some memory wastage
 4. More number of IO devices connected
- (A) Only 2 & 3
 (B) Only 2 & 4
 (C) Only 2, 3 & 4
 (D) All 1, 2, 3 & 4

- Q8** Consider a device operating on 8MBPS speed and transferring the data to memory using cycle stealing mode of DMA. If it takes 250 nanoseconds to transfer 16 bytes data to memory when it is ready/prepared. Then


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percentage of time CPU is blocked due to DMA is _____ % (rounded upto 1 decimal place)?

Q9 Consider a device operating on cycle stealing mode of DMA and transfer the data to memory in 20nanoseconds when 8 bytes data is ready or prepared. If the DMA blocks 0.1 fraction of CPU time for this transfer, then the transfer rate of the device is _____ megabytes per second?

Q10 Consider a CPU which takes 0.4 nanoseconds as interrupt overhead time when a device generates interrupt for CPU, and CPU accepts it. After that CPU takes 4 cycles to service the interrupt. The total time CPU spends for interrupt service is

1.2 nanoseconds. If the same CPU has average CPI of 4 then in the MIPS count of the CPU is _____?

Q11 A device with data transfer rate 5 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 2 CPU cycles. The byte transfer time between the device interface register and CPU or memory is negligible. The minimum performance gain of operating the device under interrupt mode over operating it under program-controlled mode is 20. The CPU has average CPI of 5, then the average instruction execution time in CPU is _____microseconds?



Answer Key

Q1 0.8~0.8

Q2 250~250

Q3 0.55~0.55

Q4 (C)

Q5 (C)

Q6 (D)

Q7 (C)

Q8 12.5~12.5

Q9 40~40

Q10 1250~1250

Q11 25~25



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Hints & Solutions

Q1 Text Solution:

Total bits sent for one character = 1 start bit + 8 character bits + 1 stop bit = 1 + 8 + 1 = 10 bits
 Efficiency of transmission line = (bits per character)/(bits transmitted per character)
 $= 8/10 = 0.8$

Q2 Text Solution:

Total bits sent for one character = 1 start bit + 8 character bits + 2 stop bits + 1 parity bit
 $= 1 + 8 + 2 + 1 = 12$ bits
 Efficiency of transmission line = (bits per character) / (bits transmitted per character)
 $= 8/12$
 $= 2/3$
 Effective transfer rate = $(2/3) * 3000 = 2000$ bits per second
 $= 2000/8$ bytes per second
 $= 250$ bytes per second

Q3 Text Solution:

CPU cycle time = $1/10\text{MHz} = 0.1$ microseconds
 Total interrupt service time = interrupt overhead time + interrupt service time
 $= 0.05 + 5 * 0.1$
 $= 0.55$

Q4 Text Solution:

Only memory is the other component of computer among all given which is directly connected to CPU. Rest all other 3 are peripheral devices and are connected with CPU using IO interface.

Q5 Text Solution:

1. Data format used in IO devices may differ from CPUs format; that's why IO interface helps in

data format conversions during data transfer between CPU & IO.

2. CPU is fastest among all components of computer.
3. Memory is faster than IO devices and slower than CPU

Q6 Text Solution:

1. ALU operation cannot be performed on IO data directly, because IO access instructions and memory access instructions in IO mapped IO are different-different and IO data can not be given to ALU directly through one instruction.
2. IO devices have their own address spaces and memory addresses are separate.
3. Less number of Instructions to access IO than the memory access instructions.
4. Less number of IO devices connected than memory mapped IO.

Q7 Text Solution:

1. ALU operation can be performed on IO data directly, because IO access instructions and memory access instructions are same.
2. IO devices don not have their own address spaces and are mapped to memory addresses only.
3. Memory wastage because the addresses which are given to IO devices cannot be used.
4. More number of IO devices connected than IO mapped IO.

Q8 Text Solution:

8 MB data preparation time in IO = 1 second
 1 byte data preparation time in IO = $1/8\text{M}$
 $= 0.125$
 microseconds
 $= 125$ nanoseconds



16 bytes data preparation time in IO = $16 * 125 = 2000$ nanoseconds

The percentage of time CPU is blocked due to DMA is = $(250 / 2000) * 100\%$

$$= 12.5\%$$

Q9 Text Solution:

Fraction of time CPU is blocked due to DMA = (transfer to memory time / preparation time)

$$0.1 = 20 / \text{preparation time}$$

preparation time = 200 nanoseconds

In 200 nanoseconds, device can prepare data = 8 bytes

In 1 nanoseconds, device can prepare data = 8 bytes / 200 nanoseconds

$$= 8000$$

bytes / 200 microseconds

$$= 40 \text{ bytes}$$

/ microseconds

In 1 second, device can prepare data = 40 bytes * 10^6 / second

$$= 40 \text{ Mbytes}$$

per second

Q10 Text Solution:

Interrupt IO time = Interrupt overhead time + Service time

$$1.2 \text{ ns} = 0.4 \text{ ns} + \text{service time}$$

$$\text{Service time} = 0.8 \text{ ns}$$

$$0.8 \text{ ns} = 4 * \text{cycle time}$$

$$\text{Cycle time} = 0.2 \text{ ns}$$

$$\text{Clock rate} = 1 / 0.2 \text{ ns} = 5 \text{ GHz}$$

$$\text{MIPS count} = \frac{5 \text{ GHz}}{4 \times 10^6} = \frac{5000}{4} = 1250$$

Q11 Text Solution:

For 5KB, device takes time = 1 second

For 1 Byte, device takes time = $1/5K = 200$ microseconds

Performance gain of interrupt IO over

$$\text{programmed IO} = \frac{\text{programmed IO time}}{\text{interrupt IO time}}$$

$$20 = \frac{200 \text{ microseconds}}{\text{interrupt IO time}}$$

Interrupt IO time = 10 microseconds

Interrupt overhead time = 2 CPU cycles = 10 microseconds

1 CPU cycle time = 5 microseconds

Average instruction execution time = CPI * Cycle time

$$= 5 * 5 \text{ microseconds}$$

$$= 25 \text{ microseconds}$$



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